Ultrasound

Exam code: 0625 & 0972

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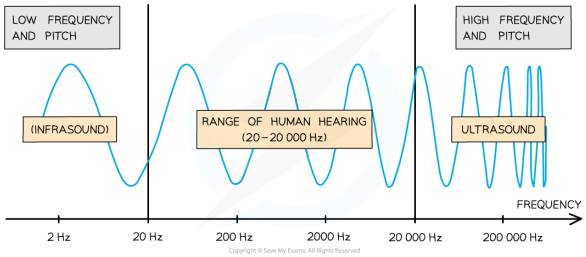
Ultrasound

Help me understand

- The approximate range of frequencies audible to humans is 20 Hz to 20 000 Hz in frequency
  - The human hearing range decreases with age

Frequencies of ultrasound



Humans can hear sounds between 20 and 20 000 Hz

- **Ultrasound** is defined as

  sound with a frequency higher than 20 kHz

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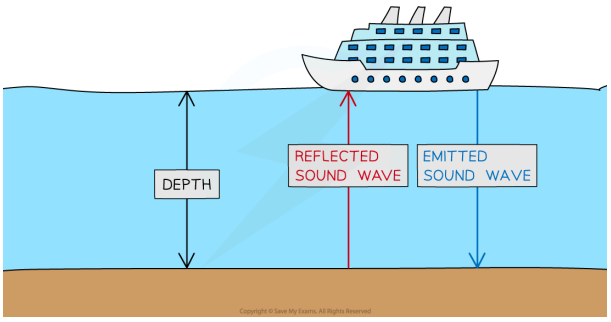
Uses of ultrasound

- non-destructive **testing of materials**
- **medical scanning** of soft tissue
- **sonar** to calculate the depth or distance from time and wave speed

Measuring the speed of ultrasound

- When ultrasound signals reach a **boundary** between two media, some of the waves are **partially reflected**
  - The remainder of the waves are **transmitted** through the material
- Ultrasound transducers:
  - **emit** and **receive** ultrasound
  - **display** the received signal on an **oscilloscope screen**
- The depth of the boundary can be determined using:
  - **Time taken** between the ultrasound being emitted and received
  - Knowing the **speed** of the ultrasound through the material

Measuring the depth of a boundary



The depth of a boundary is measured using the difference between emitted and received ultrasound pulses

- The depth of the boundary is calculated using the equation:

$$\text{depth} = \frac{1}{2} \times \text{speed} \times \text{time}$$

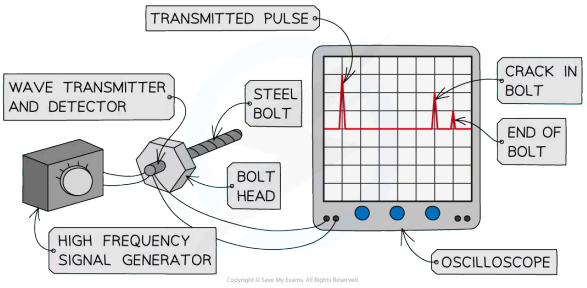
- Where:
  - depth = half the total distance travelled by the ultrasound, measured in metres (m)
  - speed = speed of the ultrasound in the medium, measured in metres per second (m/s)
  - time = total time from ultrasound emission to receipt, measured in seconds (s)

Ultrasound in material testing

- In industry, ultrasound can be used to:
  - Check for cracks inside metal objects
  - Generate images from beneath surfaces
- Ultrasound waves that reach a **crack** in an object, such as a metal bar, are reflected back from the crack earlier than

oscilloscope trace
- This allows engineers to **assess** the structure of an object

Oscilloscope display for material imperfection testing



Ultrasound is partially reflected at boundaries, so in a bolt with no internal cracks, there should only be two pulses (at the start and end of the bolt)

Ultrasound in medicine

- In medicine, ultrasound can be used:
  - To construct images of a **foetus** in the womb
  - To generate 2D images of **organs** and other internal structures (as long as they are **not** surrounded by bone)
  - As a medical **treatment** such as removing kidney stones

- The ultrasound waves are reflected back to the transducer by **boundaries** between tissues in the path of the beam
  - Examples of boundaries are between fluid and soft tissue or tissue and bone

- By taking a series of ultrasound measurements, sweeping across an area, the time measurements may be used to build up an **image**
- Unlike many other medical imaging techniques, ultrasound is **non-invasive** and is believed to be **harmless**

Ultrasound image of a baby in the womb

Sonar

- Echo sounding can be used to measure **depth** or to **detect objects underwater**
  - A sound wave can be transmitted from the surface of the water
  - The sound wave is reflected off the bottom of the ocean
- The time it takes for the sound wave to return is used to calculate the depth of the water
- The **distance** the wave travels is **twice the depth** of the ocean
  - This is the distance to the ocean floor plus the distance for the wave to return

Echo sounding

Echo sounding is used to determine water depth

Worked Example

In the diagram above, a very high-frequency sound wave is used to check for internal cracks in a large steel bolt. The oscilloscope trace shows that the bolt does have an internal crack. Each division on the oscilloscope represents a time of 0.000002 s. The speed of sound through steel is 6000 m/s.

Calculate the distance, in cm, from the head of the bolt to the internal crack.

Answer:

- Time taken between emission and detection, $t = 5 \times 0.000002 = 0.00001$ s

Step 2: Write down the equation relating speed, distance and time

$$\text{distance} = \text{speed} \times \text{time}$$

Step 3: Calculate the total distance travelled by the sound wave

$$\text{total distance} = 6000 \times 0.00001 = 0.06 \text{ m}$$

Step 4: Calculate the distance travelled by the sound wave to the crack

$$\text{distance to the crack} = 0.06 / 2 = 0.03 \text{ m}$$

Step 5: Convert the distance to cm

$$\text{distance to the crack} = 3 \text{ cm}$$

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